Von Neumann Architecture

Von Neumann Architecture

- The most widely used computer architecture model.
- Describes a system where program instructions and data share the same memory space.
    - **Single Memory**: Both data and instructions are stored in the main memory.
    - **Sequential Execution**: Instructions are fetched one at a time and executed in order.
    - **Stored Program Concept**: Instructions are represented as binary data, just like regular data.
    - **Input/Output (I/O)**: Controlled via a central bus system.

Advantages	Disadvantages
• Simplicity in design. • Easy to reprogram (change instructions in memory).	**Von Neumann bottleneck**: CPU and memory share the same bus, limiting speed when fetching instructions and data simultaneously.

Stored Program Architecture

- The underlying principle behind the Von Neumann model: a program is stored in memory alongside data, and the CPU fetches instructions from memory sequentially.
    - **Fetch**: The next instruction is fetched from memory.
    - **Decode**: The control unit interprets the instruction.
    - **Execute**: The instruction is carried out (e.g., ALU operation, data transfer).

Buses

- A bus is a communication pathway that connects different components of the computer (CPU, memory, and I/O devices).
- Consists of a set of parallel wires used for transmitting data, addresses, and control signals.
- Types of Buses:
    - **Address Bus**: Carries memory addresses from CPU to memory/I/O.
        - Unidirectional (from CPU to memory).
        - Determines the maximum memory size.
        - Example: A 32-bit address bus allows addressing up to 2^{32} GB of memory.

- **Data Bus**: Transfers actual data between CPU, memory, and I/O.
  - Bidirectional (data can move both ways).
  - Width determines how many bits can be transferred at once.
  - Example: A 64-bit data bus can transfer 64 bits in a single cycle.
- **Control Bus**: Transmits control signals to coordinate operations.
  - Read/Write signals – Indicate data transfer direction.
  - Clock signals – Synchronize components.
  - Interrupt signals – Indicate urgent CPU attention to a spontaneous event
  - Example: a 8-bit control bus can have 8 control signals

Bus Cycle

- Synchronizes data transfers between various components in a computer's system, like the CPU, memory, and other peripherals, using the system bus.
- Typically measured in MHz.
- Calculated as `Bus Cycle Time = (1 / Bus Clock Rate)`

Bus Capacity (Bandwidth)

- The maximum rate at which data can be transferred over the bus, usually measured in bytes/sec or bits/sec.
- Calculated as `Bus Capacity = Data Transfer Unit * Bus Clock Rate`